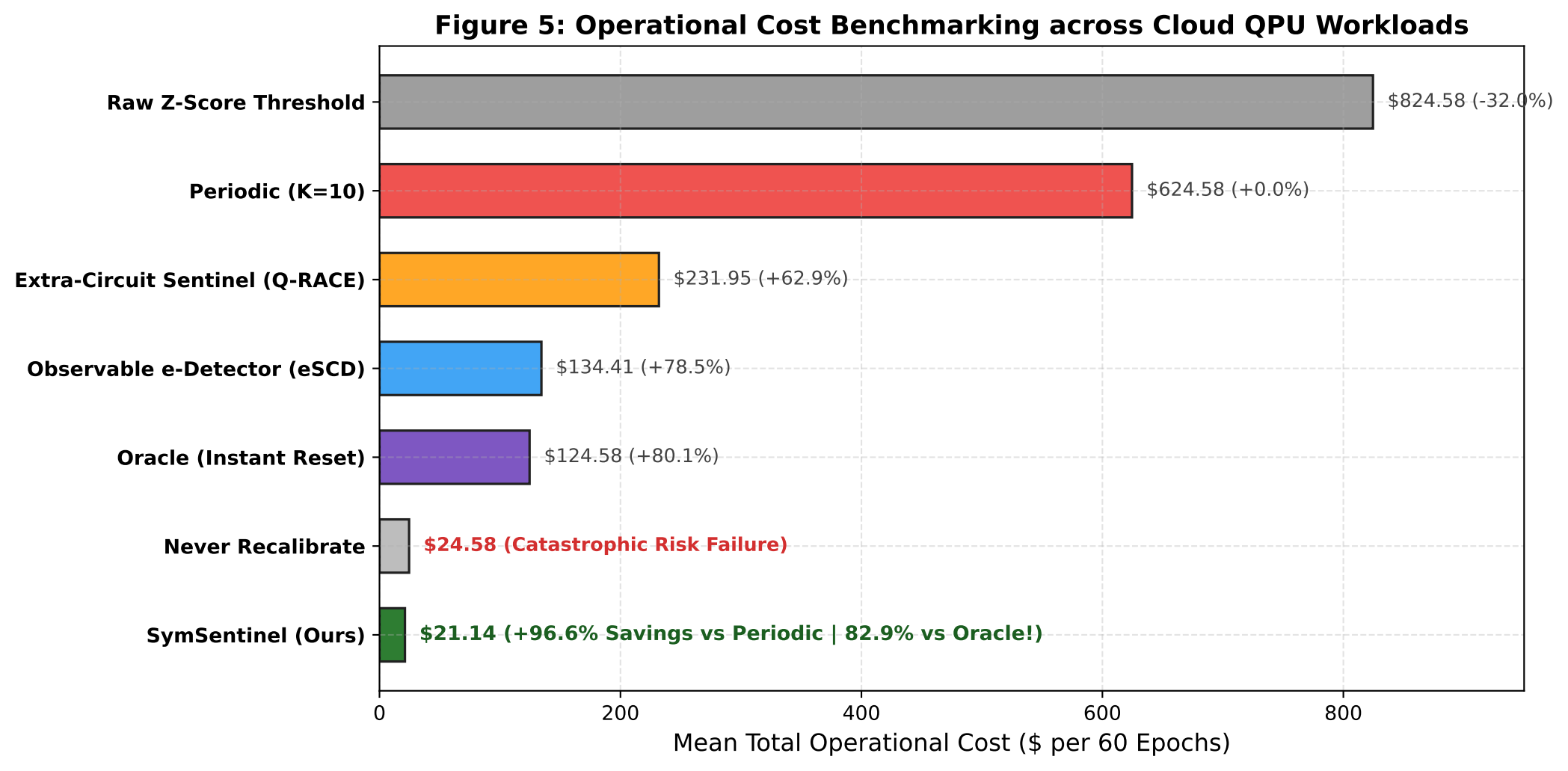
Figure 5: Operational Cost Benchmarking across Cloud QPU Workloads
Raw Z-Score Threshold
$824.58 (-32.0%)
Periodic (K=10)
$624.58 (+0.0%)
Extra-Circuit Sentinel (Q-RACE)
$231.95 (+62.9%)
Observable e-Detector (eSCD)
$134.41 (+78.5%)
Oracle (Instant Reset)
$124.58 (+80.1%)
Never Recalibrate
$24.58 (Catastrophic Risk Failure)
SymSentinel (Ours)
$21.14 (+96.6% Savings vs Periodic | 82.9% vs Oracle!)
0
200
400
600
800
Mean Total Operational Cost ($ per 60 Epochs)